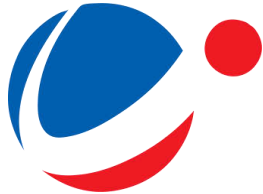


English Medium Test -8



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TEST-8

Test Booklet Series

C

GENERAL STUDIES (P) 2023 – Test – 3792

Time Allowed: Two Hours

Maximum Marks: 200

Ecology & Environment

Ecology & Environment

- Basic Concepts & Components of the Ecology and ecosystem
- Global warming, Climate change and Ozone depletion
- Environmental Issues and Pollution
- Biodiversity, flora and fauna
- Conservation efforts - India and World
- Sustainable Development
- Environmental Monitoring and Impact Assessment

+ Current Affairs (July 2022)

Sources covered

- NCERT Biology 12th – Unit 10 (chapter 13, 14, 15 & 16)
- NCERT Chemistry 11th (Part II) – Chapter 14
- ICSE Environmental Education 9th & 10th

2023

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE ASKED TO DO SO

1. Temperature and water are the most ecologically relevant environmental factors. In this context, consider the following statements:

1. Stenothermal organisms can tolerate and thrive in a wide range of temperatures.
2. Euryhaline organisms are restricted to a narrow range of salinity.

Which of the statements given above is/are **not** correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

2. With reference to the hunting of wild animals in India, consider the following statements:

1. The Chief Wildlife Wardern can permit to hunt the wild animal that has become dangerous for human life.
2. The killing of wild animal in self-defence is not considered as an offence.
3. Any wildlife animal killed or wounded in defence of a person is considered the property of Government.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

3. In the context of biodiversity, Allen's rule states that

- (a) mammals from colder climates have shorter ears and limbs to minimise heat loss.
- (b) mammals from colder climates have thick blubber of fat to protect themselves from cold.
- (c) mammals from tropical climate exhibit higher genetic diversity compared to colder climate mammals.
- (d) mammals from tropical climate have longer lifespan on an average compared to mammals from colder climate.

4. With reference to Photosynthetically Active Radiation (PAR), consider the following statements:

1. Of the total incident solar radiation more than 70 per cent of it is photosynthetically active radiation (PAR).
2. Plants on earth capture only 2-10 per cent of the total photosynthetically active radiation (PAR).

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

5. With reference to different zones of an aquatic ecosystem, consider the following statements:

1. The limnetic zone is well-lit and is dominated by plankton.
2. Profundal zone is the deep-water part of the lake or pond.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

6. Which of the following reasons account for lesser biodiversity in temperate regions as compared to tropical regions of the world?

1. Tropical regions have been subjected to frequent glaciations in the past while temperate latitudes have remained undisturbed.
2. Temperate environments are less seasonal and stable thus promoting niche specialisation.
3. There is more solar energy available in the tropics, which contributes to higher productivity.

Select the correct answer using the code given below.

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 3 only
- (d) 1, 2 and 3

7. With reference to parasitism, consider the following statements:

1. Parasites that feed on the external surface of the host organism are called endoparasites.
2. Ectoparasites live inside the host body at different sites.
3. In brood parasitism, a parasite does not feed on the organism.

Which of the statements given above is/are correct?

- (a) 3 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

8. Recently seen in the news, the 'Operation NARCOS' is a/an:

- (a) operation restricting cross border drug trade.
- (b) drive for combating cultivation of drugs in the country.
- (c) month long pan-India drive against smuggling of narcotics through rail.
- (d) awareness campaign on drug abuse among the youth.

9. Which of the following processes are involved in decomposition?

1. Fragmentation
2. Catabolism
3. Humification
4. Mineralisation

Select the correct answer using the code given below.

- (a) 1 and 4 only
- (b) 2 and 3 only
- (c) 1, 2 and 4 only
- (d) 1, 2, 3 and 4

10. This tribal community of Rajasthan are considered the first environmentalist in the world. They remove living insects from the firewood before burning. They never cut trees and wait for trees to die on their own or fall down during storms. Wildlife like Chinkara and Blackbucks move freely in their settlements.

Who among the following are being described in the above given passage?

- (a) Bhils
- (b) Gonds
- (c) Bishnois
- (d) Baigas

11. Consider the following statements:

1. The first National Wildlife Action Plan was adopted under the provisions of National Forest Policy, 1988.
2. The National Board for Wildlife is headed by the Union Minister of Environment, Forest and Climate Change.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

12. Consider the following pairs:
- | Sacred Grove | State |
|---------------------|--------------|
| 1. Than | : Assam |
| 2. Kavus | : Tamil Nadu |
| 3. Devara Kadu | : Karnataka |
- Which of the pairs given above are correctly matched?
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
13. Which of the following are the ex-situ methods of biodiversity conservation?
1. Botanical Garden
 2. Biosphere Reserve
 3. Zoo
- Select the correct answer using the code given below.
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
14. Conformers are those types of organisms that depend on external sources of body heat as they cannot regulate their body temperature as per the variations in the external temperature. In this context, which of the following are examples of conformers?
1. Shrews
 2. Hummingbirds
 3. Peacock
- Select the correct answer using the code given below.
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
15. With reference to the environmental impact of chemicals used in day-to-day life, consider the following statements:
1. Alum (aluminum sulfate) is a non-toxic liquid that is commonly used in water treatment plants to clarify drinking water.
 2. Hydrogen peroxide is a more environmentally friendly chemical for bleaching of clothes than chlorine gas.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2
16. In the context of interspecific interactions among various species, in which of the following cases only one organism is benefited?
1. Competition
 2. Predation
 3. Parasitism
 4. Commensalism
 5. Amensalism
- Select the correct answer using the code given below.
- (a) 2, 3 and 4 only
(b) 2, 4 and 5 only
(c) 1, 2, 3 and 4 only
(d) 1, 3 and 4 only
17. Recently the statue of peace, dedicated to him has been unveiled in Kashmir. He was one of the greatest flag bearers of the Vaishnav sect and propounded the doctrine of *Vishishtadvaita*. The great saint is known for scriptures like *Gita Bhasya*, in which he propagated an egalitarian society. He is believed to be the first Hindu Acharya to initiate women into *sanyas*. Identify the personality from the passage given above.
- (a) Swami Shankaracharya
(b) Swami Ramanujacharya
(c) Chaitanya Mahaprabhu
(d) Nimbarkacharya

18. Some pollutants are also divided into quantitative and qualitative pollutants on the basis of their existence in nature. In this context, which of the following are examples of qualitative pollutants?

1. DDT
2. Nitrogen oxide
3. Herbicides
4. Carbon Dioxide

Select the correct answer using the code given below.

- (a) 1 and 2 only
- (b) 1 and 3 only
- (c) 3 and 4 only
- (d) 1, 2, 3 and 4

19. With reference to Radioactive pollution, consider the following statements:

1. Radioactive dust can remain at altitudes as high as 3000 meters.
2. X-ray machines at airports are a source of radioactive pollution.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

20. With reference to the biomass production, consider the following statements:

1. Gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis.
2. Gross primary productivity minus respiration losses (R) is equal to net primary productivity (NPP).
3. The contribution of oceans to the net primary productivity of the biosphere is more than 50 percent.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 1 and 3 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

21. People for the Ethical Treatment of Animals, often in the news, is

- (a) a specialized agency of the United Nations.
- (b) an organisation under the United Nations Convention on Animal Health and Protection (UNCAHP).
- (c) an international nonprofit charitable organization.
- (d) a World Bank initiated organisation.

22. These plants were introduced into India for their lovely flowers have caused havoc by their excessive growth by causing blocks in our waterways. They are also called 'Terror of Bengal'. They grow abundantly in eutrophic water bodies, and lead to an imbalance in the ecosystem dynamics of the water body.

Which of the following plants is being described in the passage given above?

- (a) Water Hyacinth
- (b) Golden Pothos
- (c) Water Chestnut
- (d) Water Lilies

23. Which of the following statements is/are correct with reference to desert ecosystem?

1. The soil in the desert is coarse.
2. Animal diversity in the deserts is very high.
3. Plants have waxy surfaces to reduce water loss.

Select the correct answer using the code given below.

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

24. India has started several ambitious projects over the years to conserve wildlife. In this context, arrange the following conservation projects in chronological order starting from the earliest:
1. Project Tiger
 2. Project Elephant
 3. Project Crocodile
- Select the correct answer using the code given below.
- (a) 1-2-3
 - (b) 1-3-2
 - (c) 2-1-3
 - (d) 2-3-1
25. Plant biodiversity enables us to extract a wide variety of chemicals for commercial use. In this context, which of the following are extracted from plants?
1. Nicotine
 2. Caffeine
 3. Quinine
 4. Strychnine
- Select the correct answer using the code given below.
- (a) 1 and 2 only
 - (b) 3 and 4 only
 - (c) 1, 2 and 3 only
 - (d) 1, 2, 3 and 4
26. Consider the following statements with respect to the air pollution:
1. Electrostatic precipitator is used to remove particulate matter present in the exhaust from a thermal power plant.
 2. A scrubber can remove gases like nitrogen dioxide.
 3. Platinum-palladium is used in catalytic converters in the automobiles for reducing emission of poisonous gases.
- Which of the statements given above are correct?
- (a) 1 and 2 only
 - (b) 2 and 3 only
 - (c) 1 and 3 only
 - (d) 1, 2 and 3
27. Which of the following statements best describes 'Poikilotherms'?
- (a) They are organisms whose internal temperature remains constant.
 - (b) They are adopted to thrive in a wide range of natural terrain.
 - (c) They are restricted to a narrow range of latitudes.
 - (d) They are organisms whose internal temperature varies considerably.
28. Sometimes seen in the news "Lisbon Declaration" is related to:
- (a) safeguarding the outer space from space debris and junk.
 - (b) declaration signed by the countries of the European Union to stand with Ukraine in times of war and conflict.
 - (c) declaration signed at the United Nations Ocean Conference to scale up actions against maritime pollution and habitat loss.
 - (d) initiative of the World Health Organization promoting knowledge and resource sharing in times of global health crisis.
29. With reference to Eutrophication, which of the following statements is **not** correct?
- (a) It refers to the process by which a body of water becomes progressively enriched with minerals and nutrients.
 - (b) It leads to an increase in the density of animal population while restricting the growth of plants in the aquatic environment.
 - (c) It may cause loss of coral reefs due to increase in turbidity of water.
 - (d) It eventually creates a detritus layer in lakes and produces a successively shallower depth of surface water.

30. Consider the following pairs:

<i>Chemical</i>	<i>Adverse effect</i>
1. Carbaryl pesticide	: Kills honey bees
2. Toxaphene	: Deformities in the backbones of fishes.
3. Dichloro Diphenyl Trichloroethane (DDT)	: Thinning of bird egg- cells.

Which of the pairs given above is/are correctly matched?

- (a) 1, 2 and 3
- (b) 1 and 3 only
- (c) 1 only
- (d) 2 and 3 only

31. With reference to the Indian Forest Act, 1927, consider the following statements:

- 1. The State Government is empowered to notify forest land as the property of the Government.
- 2. Only the Central Government can declare the trees in a protected forest as reserved.
- 3. State Government may assign to a village community the rights of government over any land that has been constituted as a reserve forest.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

32. Primary production is defined as the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis. In this respect primary productivity depends on which of the following factors?

- 1. Plant species inhabiting a particular area.
- 2. Availability of nutrients.
- 3. Photosynthetic capacity of plants.

Select the correct answer using the code given below.

- (a) 2 only
- (b) 1, 2 and 3
- (c) 3 only
- (d) 1 and 2 only

33. Which of the following statements best describes the term POP-FAME which was seen in the news recently?

- (a) A fuel molecule developed from a bacteria by the scientific community from the United States of America.
- (b) An event organised by the Ministry of Culture appreciating the India-Korea cultural ties through means like K-pop music and drama.
- (c) An Initiative of the Ministry of Heavy Industries for faster adoption manufacturing of hybrid and electric vehicles.
- (d) A social media campaign to create awareness about incidents of racism globally.

34. With reference to the decomposition of organic matter, consider the following statements:

1. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances through a process called catabolism.
2. Decomposition rate is faster if detritus is rich in lignin and chitin.
3. The humus formed during decomposition is highly resistant to microbial action.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 1 and 3 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

35. With reference to solid waste management in India, consider the following statements:

1. In India, 'waste' is defined as any material which is broken or worn out under the Environment Protection Act, 1986.
2. Tailings are a type of e-waste material produced during recycling of discarded electronic items.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

36. Which of the following symbols/phrases are visible on the National Emblem of India?

1. Satyameva Jayate
2. Bull
3. Tiger
4. Elephant
5. Galloping Horse
6. Lion

Select the correct answer using the code given below.

- (a) 1, 2, 4, 5 and 6 only
- (b) 3, 4 and 5 only
- (c) 1, 2, 3, 4, 5 and 6
- (d) 1, 2 and 6 only

37. With reference to photochemical smog, consider the following statements:

1. It results from the action of sunlight on unsaturated hydrocarbons and nitrogen oxides.
2. It usually occurs in a cool and humid climate.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

38. Due to climate change, coral bleaching events have become a recurrent feature of late. In this context, which of the following factors can cause coral bleaching?

1. Increased exposure to ultraviolet (UV) radiation
2. Sediments such as sand covering the coral
3. Entry of excess nutrients such as ammonia and nitrate fertilizers into coral reefs
4. Excess storm water from rains entering the coral reefs
5. Salinity Changes

Select the correct answer using the code given below.

- (a) 2, 3 and 4 only
- (b) 1, 3 and 4 only
- (c) 2, 4 and 5 only
- (d) 1, 2, 3, 4 and 5

39. With reference to the National Forest Policy of 1988, consider the following statements:
1. One-fourth of the total land area of the country should be under forest cover.
 2. It provides for compensatory afforestation.
 3. Wastelands and degraded lands have been suggested to be utilized for afforestation and other plantation purposes.
- Which of the statements given above are correct?
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
40. Consider the following statements:
1. No grazing of livestock is allowed in a National Park.
 2. India's first National Park was established in West Bengal.
 3. Core zone of a Biosphere Reserve is an area of active cooperation between forest management and local community.
- Which of the statements given above are **not** correct?
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
41. Which of the following acts enacted earliest?
- (a) Air (Prevention and Control of Pollution) Act
(b) Water (Prevention and Control of Pollution) Act
(c) The Environment (Protection) Act
(d) The Forest (Conservation) Act

42. In the context of biological interaction between two or more organisms, consider the following statements regarding antibiosis:
1. None of the interactive species are benefited in antibiosis.
 2. Blue green algae that grows in ponds is associated with antibiosis interaction.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2
43. Which of the following statements is **not** correct about the fungi?
- (a) Fungi are heterotrophic organisms.
(b) All fungi are multicellular.
(c) The cell walls of fungi are composed of chitin and polysaccharides.
(d) Antibiotics like Penicillium are obtained from fungi.
44. Which of the following trees/plants are you likely to find in temperate forests?
1. Oak
 2. Maple
 3. Cottonwood
 4. Willow
- Select the correct answer using the code given below.
- (a) 1, 2 and 4 only
(b) 2 and 4 only
(c) 1 and 3 only
(d) 1, 2, 3 and 4

45. Consider the following statements with regard to Biological Oxygen Demand (BOD):

1. It depicts the amount of organic pollution present in an aquatic ecosystem.
2. High BOD level means the amount of dissolved oxygen available for marine organisms is low.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

46. Consider the following pairs:

Index/Report	Organization
1. World Cities Report	: UN-Habitat
2. Asia Pacific Sustainability Index	: UNESCAP
3. Global Findex Database report	: World Bank

Which of the pairs given above is/are correctly matched?

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1 and 3 only

47. Consider the following statements with reference to Joint Forest Management:

1. It is the model of forest management where local communities are involved in planning and conservation programme.
2. It is a statutory programme under the Indian Forest Act, 1927.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

48. Consider the following pairs:

Traditional water harvesting system	State
--	--------------

- | | |
|-----------|-------------|
| 1. Tankas | : Gujarat |
| 2. Katas | : Karnataka |
| 3. Ladakh | : Zing |

Which of the pairs given above is/are correctly matched?

- (a) 3 only
- (b) 1 and 2 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

49. Which of the following defense mechanisms are developed by plants to ward off predators?

1. Presence of fine hairs on the surface of the leaves.
2. Release of caffeine by certain plants.
3. Presence of thorns in stems and leaves.

Select the correct answer using the code given below.

- (a) 1 and 3 only
- (b) 2 and 3 only
- (c) 3 only
- (d) 1, 2 and 3

50. The term 'Evil Quartet' is frequently used to describe the threats to biodiversity. In this context, which of the following are part of Evil Quartet?

1. Climate Change
2. Over exploitation
3. Co-Extinction
4. Habitat loss and fragmentation

Select the correct answer using code given below.

- (a) 1, 2 and 3 only
- (b) 2, 3 and 4 only
- (c) 1, 2 and 4 only
- (d) 1, 2, 3 and 4

51. Consider the following:
Assertion: The Noise Pollution (Regulation and Control) Rules, 2000 have been framed under the Air (Prevention and Control of Pollution) Act, 1981.
Reason: The Air (Prevention and Control of Pollution) Act, 1981 considers noise as an air pollutant.
In the context of the statements above, which of these is true?
(a) A and R both are true, and R is the correct explanation for A
(b) A and R both are true, and R is the NOT the correct explanation for A
(c) A is correct, R is incorrect
(d) A is incorrect, R is correct
52. Which of the following are the sources of methane emissions?
1. Landfills
2. Termites
3. Natural wetlands
4. Gas drilling
Select the correct answer using the code given below.
(a) 1 and 2 only
(b) 2 and 3 only
(c) 3 and 4 only
(d) 1, 2, 3 and 4
53. Management of radioactive waste in India is governed by rules framed under which one of the following acts?
(a) Environment Protection Act, 1986
(b) Atomic Energy Act, 1962
(c) The Water (Prevention and Control of Pollution) Act, 1974
(d) Air (Prevention and Control of Pollution) Act, 1981

54. With reference to the Prime Minister Employment Guarantee Programme (PMEGP), consider the following statements:

1. It provides employment opportunities to the youth by assisting in the setting up of micro-enterprises in the non-farm sector.
2. Transgender people are considered under the special category in the PMEGP to avail higher subsidies.
3. The nodal agency to implement the PMEGP is Khadi and Village Industries Commission (KVIC).

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

55. Consider the following pairs:

<i>Initiative</i>	<i>Subject matter</i>
1. Digital India GENESIS	: National deep tech startup platform
2. India stack	: Global Repository of key projects like Aadhar and UPI
3. Meri pehchaan	: AI-led language translation platform for the transfer of knowledge and awareness

Which of the pairs given above is/are correctly matched?

- (a) 1, 2 and 3
- (b) 1 and 2 only
- (c) 1 and 3 only
- (d) 2 only

56. With reference to diversity, consider the following statements:
1. Species diversity decreases as we move away from the poles towards the equator.
 2. The Western Ghats have a greater amphibian species diversity than the Eastern Ghats.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2
57. With reference to the Environment Protection Act, 1986, the Central Government can take which of the following actions?
1. Set standards for the emission of pollutants from any source
 2. Laying down procedures and safeguards for handling hazardous substances
 3. Making rules in matters pertaining to environmental protection as well as pollution
- Select the correct answer using code given below.
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
58. Which of the following countries/organizations form a part of the I2U2 grouping?
1. Israel
 2. United States of America
 3. India
 4. United Kingdom
 5. United Arab Emirates
 6. Italy
- Select the correct answer using the code given below.
- (a) 1, 2, 3 and 5 only
(b) 1, 2, 3 and 6 only
(c) 1, 2, 4 and 6 only
(d) 3, 4 and 6 only

59. With reference to the Jagannath puri rath yatra, which one of the following statements is **not** correct?
- (a) The rath yatra is not a part of UNESCO World Intangible Cultural Heritage.
(b) The images of Gods are made of stone.
(c) Gotipua dance performance is associated with this rath yatra.
(d) It is a festival celebrated every-year during the month of Ashad.
60. Consider the following statements with reference to the Greenhouse Gases (GHGs):
1. Carbon dioxide accounts for more than fifty percent of the natural greenhouse effect.
 2. Nitrous oxide has a capacity of trapping heat 300 times more effectively than carbon dioxide.
 3. Methane gas is emitted from paddy fields and coal mines.
- Which of the statements given above are correct?
- (a) 1 and 3 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3
61. Consider the following pairs:
- | Medicinal plant | Application |
|------------------------|----------------------------------|
| 1. Karnavand | : Treatment of stomach disorders |
| 2. Sitecha Ashok | : Treatment of skin ailments |
| 3. Katal jasvand | : Cure of joint pains |
- Which of the pairs given above is/are correctly matched?
- (a) 1, 2 and 3
(b) 1 and 3 only
(c) 1 only
(d) 2 and 3 only

62. Which one of the following statements best defines the term "bioprospecting"?
- (a) Degradation of the materials into environmentally acceptable products by the action of naturally available microorganisms
 - (b) Exploring molecular, genetic and species-level diversity for products of economic importance
 - (c) Concentration of a toxin, such as pesticides, in the tissues of tolerant organisms at successively higher levels in a food chain
 - (d) Process of extraction of metals from their ores through the use of living organisms
63. In the context of the Financial Services Institutional Bureau (FSIB), consider the following statements:
- 1. It will be the sole entity for recommendations related to the appointment of the whole time Directors as well as the non-executive Chairman of the Public Sector Banks.
 - 2. A Joint Secretary of the Ministry of Finance is deemed to be the Ex-officio Chairperson of the FSIB.
 - 3. After the formation of the FSIB, all assets, interests and liabilities of the Banks Board Bureau will be transferred to the FSIB.
- Which of the statements given above is/are correct?
- (a) 1 only
 - (b) 1 and 3 only
 - (c) 2 and 3 only
 - (d) 2 only

64. Consider the following pairs:
- | National Park | State |
|------------------------------|------------------|
| 1. Tadoba National Park | : Uttarakhand |
| 2. Bandhavgarh National Park | : Madhya Pradesh |
| 3. Dudhwa National Park | : Uttar Pradesh |
- Which of the pairs given above is/are correctly matched?
- (a) 1 only
 - (b) 2 and 3 only
 - (c) 1 and 3 only
 - (d) 1, 2 and 3
65. Which of the following substances can cause depletion of the ozone layer?
- 1. Carbon tetrachloride
 - 2. Methyl bromide
 - 3. Chlorodifluoromethane
- Select the correct answer using the code given below.
- (a) 1 and 2 only
 - (b) 2 and 3 only
 - (c) 1 and 3 only
 - (d) 1, 2 and 3
66. With reference to Wildlife Protection Act, 1972, consider the following statements:
- 1. The Director of Wild Life Preservation is appointed by the State Governments.
 - 2. It bans trade and commerce in scheduled animals without a permit.
 - 3. Blue Vanda and Pitcher Plants are the only two plants to be provided protection under the act.
- Which of the statements given above is/are correct?
- (a) 1 and 2 only
 - (b) 2 only
 - (c) 3 only
 - (d) 1, 2 and 3

67. Consider the following pairs:

Pollutant	Impact
1. Lead	: Minamata Disease
2. Fluoride	: Methemoglobinemia
3. Nitrate	: Dermatitis

Which of the pairs given above is/are **not** correctly matched?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

68. Consider the following statements regarding ecological succession:

- 1. The entire sequence of communities that successively change in a given area is called sere.
- 2. Secondary succession is faster than primary succession.
- 3. Xerarch succession in plants takes place in wet areas.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

69. With reference to captive breeding, consider the following statements:

- 1. It provides a means for conserving species that may not survive in the wild.
- 2. In India, only zoos are permitted to conduct captive breeding programmes.
- 3. Nandankanan is the first-ever captive breeding centre for creating awareness of wildlife in India.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

70. Recently seen in the news "Tanner Scale" can be best described as which one of the following?

- (a) Scale to measure the progress of nations that ratified the United Nations Convention on Biological Diversity.
- (b) Scale to measure the degree of human life lost in natural disasters such as a cloud burst and landslides.
- (c) Scale describing the onset and progression of pubertal changes based on external sex characteristics.
- (d) Scale used to describe the PM 2.5 and PM 10 content in the surrounding air.

71. Coral Reefs in India are found in:

- 1. Palk Bay
- 2. Sundarbans
- 3. Lakshadweep Islands
- 4. Gulf of Kutch

Select the correct answer using the code given below.

- (a) 1, 2 and 4 only
- (b) 2 and 3 only
- (c) 1, 3 and 4 only
- (d) 1, 2, 3 and 4

72. With reference to the Indian Pharmacopoeia Commission, consider the following statements:

- 1. It is an autonomous body, fully financed by the Central Government.
- 2. Its vision is to promote the highest standards of drugs for use in humans and animals.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

73. Consider the following statements with reference to the effects of pollution on plants:

1. Hydrogen sulfide in the air causes leaf spots and defoliation.
2. High concentration of carbon monoxide causes leaf-drop, premature aging and inhibition of cellular respiration.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

74. Snow-blindness, a disease, is primarily caused by:

- (a) thinning of the ozone layer.
- (b) global warming and glacial melt.
- (c) mercury contamination in drinking water.
- (d) sudden drop in temperatures in polar regions.

75. Which of the following activities of aquatic species acts as a bioindicator of pollution?

1. Presence of Coliform bacteria
2. Abundance of diatoms and Eichhornia in water
3. Movement of Catla and Labeo fish species away from their natural habitat

Select the correct answer using the code given below.

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

76. With reference to the adaptations observed in animals, consider the following statements:

1. Mammals inhabiting colder climates have shorter ears and limbs to minimize heat loss.
2. Aquatic mammals of the polar seas have a thick layer of fat below their skin to reduce loss of body heat.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

77. NMC 2170, a term recently seen in the news is/a:

- (a) India's first indigenously developed Li-ion cell.
- (b) recently developed organic pest control mechanism by Indian Council of Agricultural Research.
- (c) modified version of CRISPR-Cas9 gene editing tool.
- (d) name of a drug to cure cancer that uses nano-materials to target the organ.

78. Which one of the following statements best defines 'endemic species'?

- (a) Species confined to a limited region for their distribution
- (b) Group of organisms that are very uncommon, scarce, or infrequently encountered
- (c) Species selected to act as an ambassador, icon or symbol for a defined habitat, issue, campaign or environmental cause
- (d) Species that arrive first in a newly created environment and through their interactions with one another, build a rather simple initial biological community

- 79.** Consider the following statements:
1. Biopiracy is the theft or usurpation of genetic materials by the patent process.
 2. The Convention on Biological Diversity gives sovereign national rights over biological resources.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

- 80.** In spring when the snow melts, water enters lakes and rivers, and the acid gets mingled into the water bodies making them highly acidic. This condition is commonly referred to as 'Spring shock' or 'Acid shock'.

Which of the following is responsible for 'Spring shock' or 'Acid shock'?

- (a) Carbonic acid
- (b) Sulphuric acid
- (c) Hydrochloric acid
- (d) Acetic acid

- 81.** Consider the following statements:

1. A pesticide is a biological or chemical agent that kills plant pests only.
2. Regular use of chemical pesticides may lead to bio-magnification.
3. BIONEEM is an environmentally friendly pesticide that leaves no residue on the food stuff.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

- 82.** Consider the following statements with respect to mulching:

1. Mulching reduces moisture loss from the soil.
2. Mulching results in an increase in the number of weeds.
3. After mulching, there is a continuous addition of organic matter to the soil.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

- 83.** With reference to the Air (Prevention and Control of Pollution) Act, 1981, consider the following statements:

1. It provides for the establishment of the Central Pollution Control Board (CPCB) for the prevention and control of air pollution.
2. Any pollutant into the atmosphere by ship or aircraft comes under the purview of the act.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

84. He was born in present-day Andhra Pradesh and later became involved in opposing the British in response to the 1882 Madras Forest Act, which effectively restricted the free movement of Adivasis (tribal communities) in their forest habitats and prevented them from practicing a traditional form of agriculture known as podu. Rising discontent towards the British led to the Rampa Rebellion of 1922, in which he played a major part as a leader. He was nicknamed "Manyam Veerudu" by local villagers for his heroic exploits.

Which of the following personalities has been described in the passage above?

- (a) Pingali Venkayya
- (b) Kaneganti Hanumanthu
- (c) Alluri Sitarama Raju
- (d) Potti Sriramulu

85. With reference to ecological pyramids, consider the following statements:

- 1. The pyramid of biomass in the sea is generally inverted.
- 2. Pyramid of energy can never be inverted.
- 3. Saprophytes are not represented in ecological pyramids.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

86. Which of the following are examples of invasive species?

- 1. Rosy wolf snail
- 2. Lantana
- 3. Carrot grass
- 4. Water hyacinth

Select the correct answer using the code given below.

- (a) 1 and 4 only
- (b) 2, 3 and 4 only
- (c) 1, 2 and 3 only
- (d) 1, 2, 3 and 4

87. Which of the following species is/are now extinct?

- 1. Steller's Sea Cow
- 2. Javan Tiger
- 3. Thylacine

Select the correct answer using the code given below.

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

88. Which of the following countries participated in the recently held Caspian Summit, 2022?

- 1. Russia
- 2. Iran
- 3. Uzbekistan
- 4. Kazakhstan
- 5. Turkmenistan
- 6. India

Select the correct answer using the code given below.

- (a) 1, 2, 3, 4 and 6 only
- (b) 2, 4, 5 and 6 only
- (c) 1, 2, 4 and 5 only
- (d) 1, 3, 5 and 6 only

89. Recently ISRO has achieved the successful launch of the PSLV Orbital Experimental Module (POEM). In the context of this development, consider the following statements:

- 1. POEM allows in-orbit scientific experiments using the third stage of the Polar Satellite Launch Vehicle.
- 2. POEM is devoid of any Li-ion batteries and derives all the power from solar energy, making it a green device.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

90. Which of the following are used as biopesticides in controlling plant pests?
1. Pyrethrum
 2. Bacillus Thuringiensis
 3. Rotenone
- Select the correct answer using the code given below.
- (a) 1 and 2 only
 - (b) 2 and 3 only
 - (c) 1 and 3 only
 - (d) 1, 2 and 3
91. Consider the following statements with reference to the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES):
1. It was adopted as a result of the resolution adopted at the meeting of the International Union for Conservation of Nature.
 2. It is legally binding on the parties.
 3. Export or import of any species mentioned in Appendices of CITES require a prior grant of a permit.
- Which of the statements given above are correct?
- (a) 1 and 2 only
 - (b) 2 and 3 only
 - (c) 1 and 3 only
 - (d) 1, 2 and 3
92. With reference to Mission Shakti of the Ministry of Women and Child Development, consider the following statements:
1. All the sub-schemes under the Mission Shakti are funded in a 60:40 ratio between the Centre and the states respectively.
 2. The Beti Bachao Beti Padhao has become a part of the Mission Shakti.
- Which of the statements given above is/are correct?
- (a) 1 only
 - (b) 2 only
 - (c) 1, 2 and 3
 - (d) 1 and 3 only

93. The term 'PIVOT' recently in the news is related to a/an
- (a) counter-corruption mechanism developed and adopted during the latest BRICS Summit.
 - (b) an AI-based tool to detect cancer-causing genes.
 - (c) recent developments observed through the James Webb Telescope.
 - (d) term proposed by Prime Minister of India to reflect upon India's stand against terrorism.
94. Living organisms over time have acquired various mechanisms to deal with harsh abiotic conditions in their habitat. In this context, consider the following pairs:
- | Activity | | Organism |
|-----------------|---|-----------------|
| 1. Hibernation | : | Bear |
| 2. Aestivation | : | Snail |
| 3. Diapause | : | Zooplankton |
- Which of the pairs given above is/are correctly matched?
- (a) 1 only
 - (b) 1 and 2 only
 - (c) 2 and 3 only
 - (d) 1, 2 and 3
95. Recently seen in the news, the treeshrew is a fossil of a:
- (a) plant
 - (b) reptile
 - (c) bird
 - (d) mammal
96. Which of the following can help in promoting sustainable development?
1. Use of Non-conventional Sources of Energy
 2. Use of Compressed Natural Gas (CNG)
 3. Biocomposting
 4. Solar Power through photovoltaic cells
- Select the correct answer using the code given below.
- (a) 1 and 2 only
 - (b) 1, 2 and 3 only
 - (c) 3 and 4 only
 - (d) 1, 2, 3 and 4

97. To cope with extreme conditions, desert plants have adopted various mechanisms to minimize water loss. In this context, which of the following adaptations can be observed in desert plants?

1. Arrangement of stomata in deep pits
2. Modification of leaves into spines
3. Photosynthesis by stems

Select the correct answer using the code given below.

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

98. Often seen in the news, the Montreal Protocol is related to

- (a) prevention of pollution of the marine environment by ships.
- (b) controlling the emission of ozone depleting substances.
- (c) protection of water resources that are stressed by overuse and pollution.
- (d) protection of endangered species listed in the IUCN Red List.

99. With reference to standing crop, consider the following statements:

1. It refers to the total mass of producers or autotrophs in a food chain at a particular time.
2. It is measured as the mass of living organisms or the number in a unit area.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

100. In a given food chain of a terrestrial ecosystem, an animal which is strictly herbivore will necessarily be classified as a:

- (a) primary producer
- (b) tertiary consumer
- (c) secondary consumer
- (d) primary consumer